Email: **rd.rontu1711@gmail.com**
rontu.das@prescience.in
Contact no. **(+91) 6290580486**
ORCID ID: **0000-0002-7129-999X**
Scopus ID: **58506803100**
Web of Science ID: **PMR-0499-2026**
Google Scholar ID:
https://scholar.google.co.in/citations?user=oTjA6wcAAAAJ&hl=en&oi=ao
***h*-index: 6, *i10*-index: 5**

RONTU DAS

website: rontu1997.github.io
Research Scientist,
Prescience Insillico Pvt. Ltd.
Address: 66/111 P. Road, Belgachia,
Howrah.
Pin code - 711108.



Curriculum Vitae

Present Position

Research Scientist at Prescience Insillico Pvt. Ltd., developing AI-agent-driven scientific computing solutions for industrial R&D. I translate client-specific scientific challenges into computational workflows and build AI systems tailored to practical applications using Microsoft Discovery and Microsoft Foundry.

Professional Summary

Research Scientist with a Ph.D. in Chemical Engineering, specializing in AI-agent-driven scientific computing for industrial R&D. Experienced in translating client-specific scientific challenges into computational solutions through scientific problem formulation, AI engineering, and workflow development. Develops AI agents using Microsoft Discovery and extends their capabilities through Microsoft Foundry, integrating knowledge retrieval, memory, guardrails, and scientific-tool orchestration. Connects AI agents with scientific applications through Docker-based cloud workflows and Model Context Protocol (MCP) servers for local execution. Combines expertise in Molecular Dynamics and computational materials science with AI engineering to develop reliable, scientifically validated computational workflows.

Ph.D. thesis under the supervision of Dr. Debashis Kundu
Title of the thesis: Artificial Intelligence Assisted Molecular Dynamics Simulation of Carboxymethyl Cellulose–Polyvinyl Alcohol Click Hydrogels for Various Applications
Master of Technology thesis under the supervision of Dr. Debajyoti Goswami
Title of the thesis: Biocatalytic Production of Essential Fatty Acids in Ionic Liquid/Surfactant Mixed System

Copyright Granted (for COMPUTER SOFTWARE)

1.	**Title**	**NAMoSTe (Novel Atomic & Molecular Structure Explorer)**		
	Application Number	29693/2024-CO/SW	**Certificate No.**	SW-20338/2025
	Date of Filing	23-09-2024	**Date of Granted**	04-04-2025
2.	**Title**	**Swelling analysis for Materials (SaMs)**		
	Application Number	6058/2025-CO/SW	**Certificate No.**	SW-2025021043
	Date of Filing	21-02-2025	**Date of Granted**	23-07-2025
3.	**Title**	**Ionic Conductivity Simulator (IConS)**		
	Application Number	6063/2025-CO/SW	**Certificate No.**	SW-2025021136
	Date of Filing	21-02-2025	**Date of Granted**	04-08-2025

Email: **rd.rontu1711@gmail.com**
rontu.das@prescience.in

RONTU DAS
rontu1997.github.io

Key Research Areas: Thermodynamics, Molecular Dynamics Simulation, Quantum Chemical Calculation, Data-Driven Neural Network, Physics-Informed Neural Network, Polymeric gel, Material Discovery

***Research Publications*:**

Refereed Journals	
2026	
13	***Rontu Das***, Debashis Kundu. Bridging Atomistic Simulations and Reservoir Computing for Predicting Structural and Transport Properties of Thiol–Ene Click-Cross-Linked Carboxymethyl Cellulose Hydrogels. **The Journal of Physical Chemistry B** 130(11) (2026) 3167-3186. DOI: 10.1021/acs.jpcb.5c08098
12.	***Rontu Das***, Debashis Kundu. Understanding the Molecular Level Interaction of Lithium-ions with Cellulose Based Eutectogel: A Molecular Dynamics Study. **Advanced Theory and Simulations** (2026). DOI: 10.1002/adts.202501934
11.	***Rontu Das***, Debashis Kundu. Molecular dynamics simulation of thiol-ene 'click' copolymerized and blended carboxymethyl cellulose and 4-arm polyethylene glycol hydrogel used as ciprofloxacin drug carrier. **Chemical Engineering Science** 324 (2026) 123356. DOI: 10.1016/j.ces.2026.123356
2025	
10.	Ayush S. Gulhane, ***Rontu Das***, Navnath T. Hatvate and Debashis Kundu. Optimizing Solvent Systems for Effective Lutein Solvation: A Molecular Dynamics Approach Using 2-Methyl Tetrahydrofuran, Deep Eutectic Solvent, Microemulsion, and Hexane. **Macromolecular Theory and Simulations (2025).** DOI:10.1002/mats.202500016
9.	***Rontu Das*** and Debashis Kundu. Adsorption of Methylene Blue Dye in the Thiol-Norbornene Crosslinked Polyvinyl Alcohol Hydrogel Surface: A Molecular Dynamics Study. **Environmental Science and Pollution Research** (2025). DOI: 10.1007/s11356-025-36309-3
8.	Ritesh K. Patil, ***Rontu Das***, Debashis Kundu. Structure and Dynamics of Anacardic Acid in Hexane, Ethanol and Carbon Tetrachloride: A Molecular Dynamics Investigation. **Journal of Molecular Modelling** (2025) 31, 107. DOI: 10.1007/s00894-025-06320-z
7.	Abhinav Tidke, ***Rontu Das***, Debashis Kundu. Unveiling Molecular Interactions in Chocolate by Molecular Dynamics: A Study on Caffeine, Fatty Acids and Sugars. **Journal of Chemical & Engineering Data** (2025). DOI: 10.1021/acs.jced.4c00665
6.	***Rontu Das***, Debashis Kundu. Enlightenment the Dynamic Behavior of Norbornene–Modified 'Click' 4–arm Polyethylene Glycol Hydrogel: Delving into Framework Properties and Transport Properties through Molecular Dynamics Simulations. **Computational Materials Science** (2025) 247, 113516. DOI: 10.1016/j.commatsci.2024.113516
2024	
5.	***Rontu Das*** and Debashis Kundu. Understanding the Supramolecular Chemistry of Lutein with Gelatin,

	Octenyl Succinic Anhydride-Modified Starch Polymers in Simulated Gastro-Intestinal Buffers: A Molecular Dynamics Study. **Macromolecular Chemistry and Physics** (2024) 2400295. DOI: 10.1002/macp.202400295
4.	Abhinav Tidke, ***Rontu Das***, Uday S. Annapure, Debashis Kundu. Solvent Optimization for Caffeine and Tannic Acid Extraction from Guarana Seed: Insights from Molecular Dynamics Simulations. **Journal of Molecular Liquids** (2024) 416, 126462. DOI: 10.1016/j.molliq.2024.126462
3.	Aradhana V. Thombre, ***Rontu Das***, Debashis Kundu. Lithium Ion Transportation in Lignin Infused Polyvinyl Alcohol Based Eutectogel: A Molecular Dynamics Framework. **Journal of Energy Storage** (2024) 102, 114060. DOI: 10.1016/j.est.2024.114060
2.	Shreyash P. Deshmukh, ***Rontu Das***, Debashis Kundu. Unraveling the underlying structural & transport mechanism of lithium-ion within Lithium bis(trifluoromethanesulfonyl)imide subjected to organic & inorganic matrix based Eutectogel. **Journal of Power Sources** (2024) 600, 234270. DOI: 10.1016/j.jpowsour.2024.234270
	2023
1.	***Rontu Das***, Debashis Kundu. Structural and Transport Properties of Norbornene-Functionalized Poly(vinyl alcohol) "Click" Hydrogel: A Molecular Dynamics Study, **ACS Sustainable Chemistry & Engineering** (2023) 11(29) 10812–10824. DOI: 10.1021/acssuschemeng.3c01948

Academic Records

Ph.D. in Chemical Engineering	Institute of Chemical Technology	Awarded	2021 – 2026
Master of Technology in Chemical Engineering (Specialization: Petroleum Refinery Engineering)	University of Calcutta	8.07 (CGPA)	2019 – 21
Bachelor of Chemical Engineering	Maulana Abul Kalam Azad University of Technology, West Bengal	8.74 (DGPA)	2015 – 19
12th Std.	Bantra Madhusudan Palchowdhury High School, West Bengal	78.00 %	2013 – 15
10th Std.	Bantra Madhusudan Palchowdhury High School, West Bengal	76.00 %	2013

Achievements

- Awarded the highly competitive **CSIR-Senior Research Fellowship (Direct)** by the Council of Scientific & Industrial Research (May 2024).
- **Demonstrated** strong technical aptitude by successfully qualifying the Graduate Aptitude Test in Engineering (GATE) in Chemical Engineering (2020).

Professional Experience

Email: **rd.rontu1711@gmail.com**
rontu.das@prescience.in

RONTU DAS
rontu1997.github.io

- **Research Scientist | Prescience Insillico Pvt. Ltd. | Jul 2026 – Current**
  - **AI-Agent Development:** Develop AI-agent-driven scientific computing solutions for industrial R&D by translating client-specific scientific challenges and research requirements into practical computational workflows. Build agents using Microsoft Discovery and extend them through Microsoft Foundry with knowledge bases, web search, memory, guardrails, scientific tools, and workflow orchestration for reliable multi-step tasks.
  - **Cloud and Local Integration:** Containerize scientific software with Docker and publish images to Microsoft Azure Container Registry for cloud deployment and integration. Develop Model Context Protocol (MCP) servers for local workflows, enabling agents to access scientific software through standardized interfaces and execute specific computational tasks.
  - **Research Approach:** Develop AI agents that translate research objectives expressed in natural language into structured workflows linking knowledge retrieval, scientific tool selection, input preparation, simulation execution, and result analysis. My simulation research showed that reliable use of scientific software also requires understanding numerical methods, models, parameters, boundary conditions, software configuration, and data analysis; I therefore prioritize scientific validation and expert judgment. This work builds on my academic foundation in Chemical Engineering and computational materials research.

Research Experiences

- **CSIR-Senior Research Fellowship (Direct) | May 2024 – Apr 2026**
  - **Project:** Amalgamation of Molecular Dynamics and Machine Learning Strategies for the Creation of Supersorbent Thiol-ene 'Click' Hydrogel
  - **Research Focus:** AI Assisted Molecular Dynamics Simulation for thiol-ene 'click hydrogel for predicting long term trajectory
- **Junior Research Fellow | SERB-Sponsored Project | Dec 2023 – Apr 2024**
  - **Project:** Artificial Intelligence and Molecular Dynamics Driven Modeling Platform for 'Click' Hydrogel
  - **Research Focus:** Engineered an integrated computational platform combining Artificial Intelligence (AI) and Molecular Dynamics (MD) simulations to analyse and predict the structural and functional behaviours of 'click' hydrogels.
- **Junior Research Fellow | DST-Sponsored Project |Sept 2021 – Dec 2023**
  - **Project:** Sustainable, Biodegradable, and Affordable Substitutes for 'Single-Use Plastics'
  - **Research Focus:** Researched and developed eco-friendly, biodegradable material alternatives to single-use plastics by formulating novel composites using castor oil and agricultural stubble aggregates.